Countersign: On-Device Detection of Impersonation Scam Calls with Family-Linked Response

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Team *losers*, RAKSHAM 2026 Finale, Track 1 (voice-cloning fraud)
Code, data cards and every result: https://github.com/dope-XD/countersign

***Abstract*—Impersonation calls ("digital arrest", fake bank KYC, a relative in trouble) persuade victims to transfer money while the call is still in progress. We present Countersign, an Android application that detects such calls on the victim's own phone and brings in the victim's family before money moves. Speech is transcribed on the device, and two independent judges assess the conversation: a multilingual encoder (e5-small) fine-tuned on 267,601 conversation windows, which scores the call as a whole, and a negation-aware rule base of 19 scam "moves" that supplies the explanation shown to the user. A warning is raised when either judge is confident; the family alarm requires two independent sources. A separate check asks a family member's own phone whether it placed the call, which defeats voice cloning and caller-ID spoofing without using audio. On 1,019 genuine bank conversations never used in training, 2.3% raise a warning (a keyword baseline: 50.1%); 87.8% of 4,499 held-out scam conversations do. We report the weaknesses we measured, among them scripted English scams and talk about scams.**

***Index Terms*—scam detection, social engineering, on-device inference, speech, multilingual, privacy**

I. Introduction

In a typical "digital arrest" call, a caller claiming to be from the CBI tells the victim that a parcel in her name contained drugs, keeps her on the line for an hour, forbids her to tell anyone, and finally asks her to move her savings to a "safe account" to clear her name. Fake KYC updates and "your son has had an accident" calls follow the same script. Two properties make these calls hard to stop: the harm happens *during* the call, and the victim is deliberately isolated from the people who would recognise the scam.

A detector for this setting must therefore work in real time, on the victim's phone, in the languages the call is held in (English, Hindi and their mixture, Hinglish), and it must hand the decision to a person the victim trusts. It must also not alarm on genuine calls: a real bank's fraud team uses the same vocabulary (*OTP*, *account*, *police*) as the scammer. Our first prototype, which scored keyword cues, warned on half of genuine bank calls (Table II); this observation shaped the design below.

Contributions.

- An on-device detection pipeline that judges *what the caller is doing* across the whole conversation, using a fine-tuned multilingual encoder, combined with interpretable cue rules through a two-source rule for the highest alert level (Section IV).
- A caller check that needs no audio: the claimed family member's phone reports its own calls, so a cloned voice or a spoofed number fails because that phone never placed the call (Section V).
- A response path that reaches the family (full-screen alarm, SMS fallback, escalation) and a two-tap recovery from false alarms, with privacy constraints enforced in the architecture (Sections V, VI).
- An evaluation on data the models never saw, with false-positive rates and named failure cases (Section VIII); all numbers are reproducible from the repository.

II. Related Work

Deployed on-device detectors. Google's Scam Detection analyses calls on Pixel phones for "patterns commonly associated with scams" and keeps no audio or transcript [10]. It is English-only, limited to specific handsets, and does not involve the victim's family.

Models for conversational scams. PreScam [1] collects 11,573 scam conversations from user reports and finds that a fine-tuned BERT predicts scam progression better than zero-shot large language models (83.4 vs. 63.3–69.9 AUC). SCRIPTMIND [3] models scams as crime scripts with fine-tuned small language models, and Shen et al. [4] study real-time LLM warnings and the trade-off between timeliness and precision. These results motivate our choice of a compact fine-tuned encoder over an on-phone generative model.

Data. ICFD-31k [2] provides about 31,000 bilingual (English and Hinglish) bank-call conversations, synthetic and generated with Llama 3.3 70B, with streaming verdicts every 3 s, ten fraud families in the main set and five unseen fraud types in a separate cross-domain set. It is our main training and test source.

Synthetic voice detection. Leaderboards of speech deepfake detectors report that small models trade accuracy for efficiency [8]. We use the first layers of a post-trained AntiDeepfake model [7] and, given its error rate over phone audio (Section VIII), treat it as weak evidence only.

III. System Overview

Fig. 1 shows the three parties. Every family member installs the app. The *protected user's phone* captures the call, transcribes it and decides; the *relay*, a small server without models, links family phones and carries call events, caller checks and alerts as identifiers; an optional *detection server* gives a second opinion on calls the phone already finds suspicious.

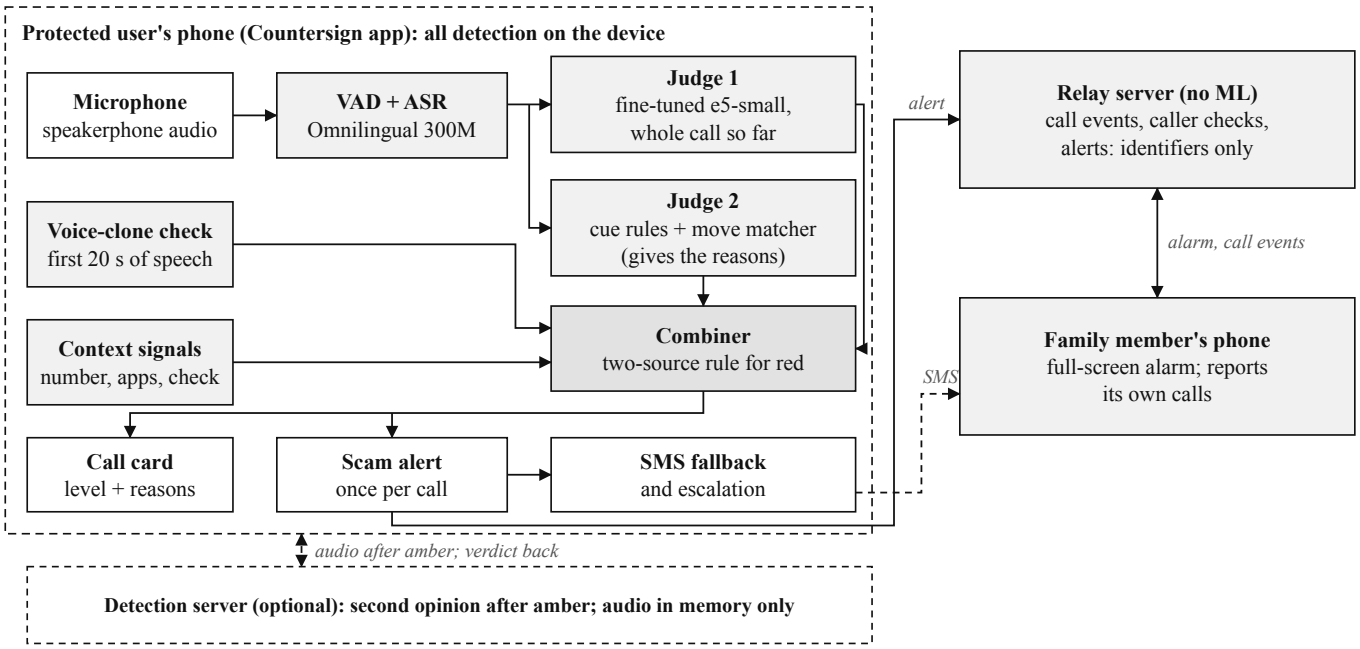


Fig. 1. Architecture. All listening and judging happen on the protected user's phone. The relay carries identifiers only; the detection server is optional and receives audio only after the phone itself has raised a warning.

Threat model. The attacker calls from an unknown number or spoofs a known one, impersonates an authority, a bank or a family member, may use a cloned voice, and steers the victim towards a transfer, a one-time password, or installing a remote-access app. We assume the attacker cannot control the family member's phone. Detection requires the call to be on speaker (Section IX).

Alert levels. A call is at one of four levels: *none*, WATCH (a few signs; no visible warning), AMBER (a warning card naming the scam and the reasons) and RED ("This is likely a scam. Hang up." and the family alarm). Within a call the level never decreases, except when the user declares the call genuine.

IV. On-Device Detection

A. Audio capture and speech recognition

Android lets an accessibility service keep the microphone during a call; the app uses one to record 16 kHz mono audio into memory while a call is active and listening is enabled. Silero VAD [9] segments speech (threshold 0.5, minimum silence 0.4 s, segments up to 12 s). Each segment is transcribed by Meta's Omnilingual ASR [6] (300M parameters, CTC, int8) through sherpa-onnx, and its audio is discarded. We selected this recogniser on our 59 spoken calls: its word error rate was 28%, against 65% and 44% for the Hindi and English IndicConformer models, each of which also loses the other language of a code-mixed call.

B. Judge 1: the conversation model

The first judge reads the conversation so far and estimates whether the caller has made a scam move.

Model. We fine-tune `multilingual-e5-small` [5] (118M parameters) with mean pooling and a logistic head. Inputs are the most recent part of the call (left-truncated to 320 tokens, which covers 99% of 1,200-character windows), normalised but kept in its original script, with the e5 "query:" prefix.

Training data. ICFD-31k streaming windows plus our own sets (Hinglish call lines, public scam dialogues and SMS corpora, and template-generated genuine calls that use scam vocabulary), 267,601 windows in total with 24.6% positive. Our own rows are repeated eight times to balance the synthetic majority. We train for one epoch with AdamW and a linear schedule with warm-up, and select the checkpoint on the development set only (mean per-source AUC 0.989).

Splits. ICFD's validation and test splits contain only fraud, so we hold out conversations of its training split by a hash of the conversation id: 10% for development and 10% as the only ICFD test set with genuine calls. ICFD's own test split and its cross-domain split stay untouched.

On the phone. The model runs as an fp32 ONNX graph (470 MB) with onnxruntime; it reproduces PyTorch to within 10^{-5}, whereas int8 quantisation shifted scores by up to 0.20 and was rejected. Tokenisation uses our own Kotlin implementation of the SentencePiece unigram model, identical to the reference tokenizer on 3,005 test texts.

Accumulation. Let p_t be the model's score after segment t and $s_t = \frac{1}{2}(p_t + p_{t-1})$. The call reaches

$$\text{level}_t = \begin{cases} \text{RED} & \text{if } s_t \geq 0.9855 \text{ and } s_{t-1} \geq 0.9855, \\ \text{AMBER} & \text{if } s_t \geq 0.142, \\ \text{WATCH} & \text{if } s_t \geq 0.0102, \\ \text{none} & \text{otherwise,} \end{cases} \tag{1}$$

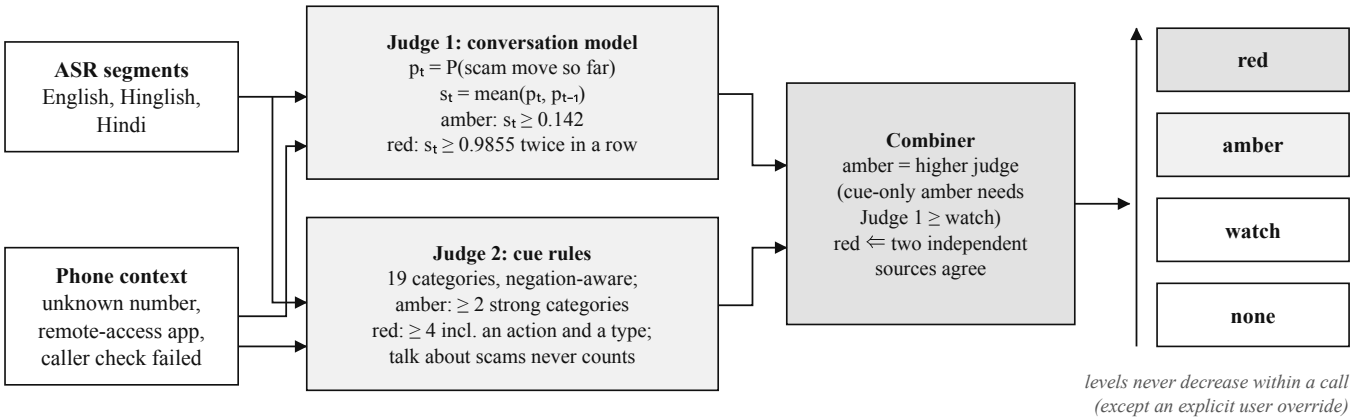


Fig. 2. Decision. Both judges see the transcript and the phone's context. AMBER follows the more confident judge; RED requires two independent sources.

and the level of the call is the highest level_t reached so far. The thresholds were set on 1,788 genuine development calls at 10%, 3% and 0.5% false-positive rate. A known contact or a verified caller raises them (0.142, 0.9865, 0.9932). A failed caller check or an opened remote-access app is sufficient for AMBER on its own. Together with AMBER from speech, these signals, an unknown video call or an international number produce RED.

Reasons. A move matcher embeds each segment with the base e5 model and compares it with written descriptions of 17 scam moves (e.g. "claims to be police", "asks to keep it secret", "asks for the OTP"). Each move has its own threshold, the 99.5th percentile of similarity on held-out genuine turns. A move is reported only when p_t rose by at least 0.15 or the level changed. The matcher never sets the level.

C. Judge 2: cue rules

The second judge is a rule base of phrases in English, Hinglish and Hindi grouped into 19 categories (authority, threat, secrecy, money request, credentials, remote access, urgency, and others). A negation window prevents "we will never ask for your OTP" from counting as a request; a *discussion* category (talk about scams) never counts towards RED. Combinations of categories name the scam type (e.g. *digital arrest* = authority plus a threat, secrecy or a video-call demand). The rules reach WATCH with one category, AMBER with two strong categories, and RED with four including an action and a type. A context signal counts once: it is never also used as a cue towards the same AMBER.

D. Context and voice evidence

The phone contributes signals of its own: an unknown or international number, a remote-access app (AnyDesk, TeamViewer and others) or a payment app brought to the front during the call, a call longer than 15 minutes, a failed caller check, the user reading out four to eight digits on an unknown call, and an unknown number claiming to be a bank or authority. A voice-clone check scores the first 20 s of speech with the first six layers of AntiDeepfake wav2vec-large [7] and a logistic head (int8, 127 MB). Because of its error rate over phone audio (Table IV), a synthetic-voice signal can raise WATCH to AMBER but never produces RED.

E. Combination

Fig. 2 summarises the rule. AMBER is the higher of the two judges, except that cue words alone do not produce AMBER while the conversation model is below WATCH: genuine bank and courier calls use the same words. RED requires two independent sources: one judge at RED from speech together with the other judge at AMBER or above; a judge that already paired speech with a strong phone signal; the user pressing "Alert family"; or the detection server. The server, when present, can raise the level but never lower it.

F. Optional second opinion

Once the phone itself is at AMBER, it may stream the call to the detection server, which runs a larger recogniser (Omnilingual 1B), the same conversation model, and the full 24-layer AntiDeepfake model on 4 s windows. Audio is kept in a rolling 20 s buffer in memory and dropped when the stream closes.

V. Caller Check and Response

Caller check. When a caller claims to be a family member, or calls from a family member's number, the relay compares the claim with that member's own reported call events around the moment the call rang (Fig. 3). An outgoing call to the protected user verifies the caller; no call while the member's app was seen in the last two minutes yields "Not Rahul's phone"; an app not seen for two minutes yields "could not check", never a false negative. Negative verdicts wait up to 6 s for late events. Numbers are matched to member identifiers on the phone and never leave it. The user may also ring the member, whose phone asks "Is it you?"; a "yes" is signed by a hardware-backed key after biometric confirmation.

Alert and escalation. The first time a call reaches RED, the phone sends one alert carrying the call identifier, level, scam

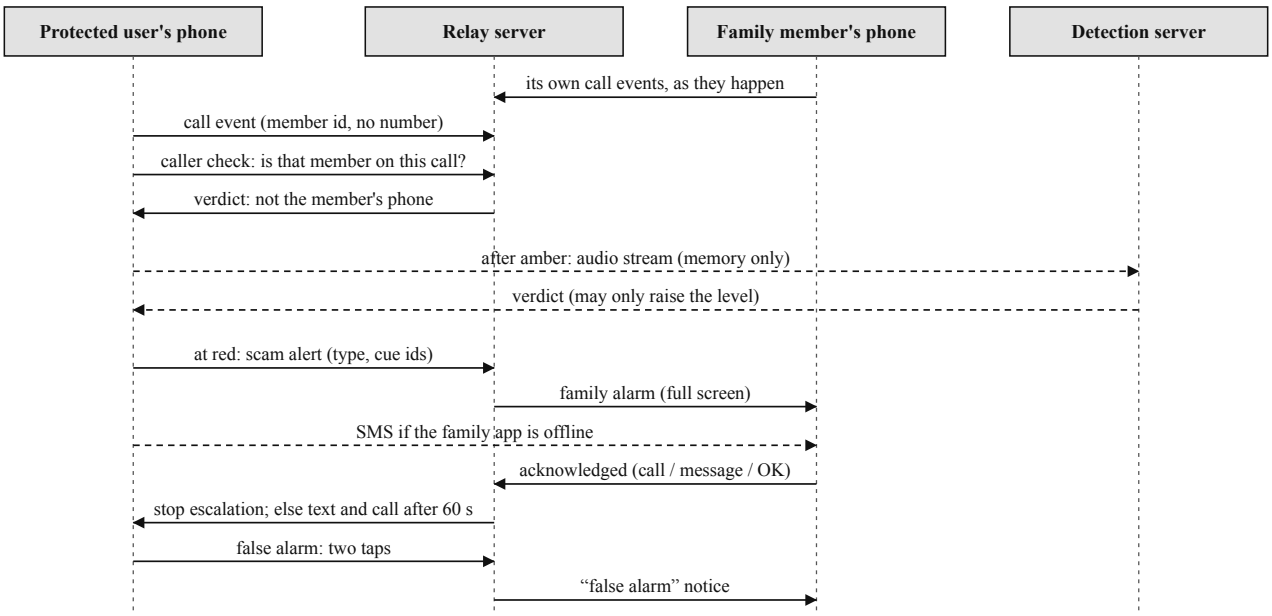


Fig. 3. Message flow for one scam call: caller check, optional second opinion, alert, family alarm with SMS fallback and escalation, and the false-alarm path. Dashed: optional or fallback paths.

TABLE I
WHERE EACH KIND OF DATA GOES

Data	Handling
Call audio	Phone memory, discarded per segment; to the server only after AMBER, in memory; one 20 s AES-256 encrypted clip for a confirmed scam, deleted after 30 days
Transcript	Phone memory for the duration of the call; never stored or sent
Phone numbers	On the phone only; matched to member identifiers by the last ten digits
Relay	Call events (member ids), check verdicts, alert (level, type, cue ids)
Logs	Levels, types and cue identifiers; never words or numbers

type and cue identifiers. The relay rings every family member whose app is online with a full-screen alarm (“Mummy may be on a scam call”, with *Call* and *Message* actions) and returns the offline members, whom the phone texts from its own SIM. If nobody acknowledges within 60 s, the phone texts all members and places a call to the first one.

Recovering from false alarms. “I know this caller” or “It’s not a scam”, confirmed once, stops warnings for the call, closes any server stream, clears the alarm on family phones (“False alarm: Mummy says it’s fine”) and stores one identifier-only record for human review.

VI. PRIVACY

Table I lists every data flow. Detection, the explanation and the caller check work without sending speech or text anywhere; the relay cannot reconstruct what was said. “Check a recording or message” applies the same analyser to a file or pasted text, entirely on the phone.

VII. IMPLEMENTATION

The Android client is written in Kotlin; the server, the training code and the evaluation are in Python. Every decision component exists in both languages, and continuous integration checks that both give identical results on shared test vectors (tokenizer, conversation accumulator, cue rules, combination). The model pack is a single 798 MB download, verified by hash, sized for a phone with 6 GB of RAM; detection works offline after the download. Of the product code, 66% was written during the 48-hour finale (measured with `git blame` against the check-in commit).

VIII. EVALUATION

Protocol. Thresholds were chosen on development data only. Test sets were never used for training, threshold selection or checkpoint choice. A call counts as *flagged* when it reaches AMBER or RED at any point. ICFD conversations are fed in their streaming chunks, our calls segment by segment as the phone does.

False alarms on genuine calls. Table II shows the effect of judging the conversation instead of keywords: on clear genuine bank calls, warnings fall from 50.1% to 2.3% and alarms from 8.9% to 0.4%. The keyword baseline catches more scams only because it flags most genuine calls as well.

Scams. 87.8% of scam conversations in ICFD’s test split reach AMBER and 59.9% reach RED, half a minute to a minute into the call. Detection drops on the five fraud types absent

TABLE II
ICFD-31K CONVERSATIONS NOT USED IN TRAINING: SHARE REACHING AMBER / RED

Set	n	Countersign	Keyword baseline
Genuine, clear	1,019	2.3 / 0.4	50.1 / 8.9
Genuine, ambiguous	511	4.5 / 0.8	70.8 / 15.5
Scam, test split	4,499	87.8 / 59.9	94.3 / 51.5
Scam, held out (clear)	193	99.5 / 82.9	–
Scam, held out (subtle)	370	98.9 / 75.9	–
Unseen types (clear)	760	75.8 / 42.9	90.9 / 34.7
Unseen types (subtle)	240	54.6 / 17.5	78.3 / 22.5

Percent. Median time to AMBER: 45 s (test split), 36 s (held out).

TABLE III
OTHER HELD-OUT SETS (TEXT, FED TURN BY TURN)

Set	n (scam / genuine)	Scams	Genuine
Team calls	16	13	93.8%	7.7%
Adversarial rewrites	16	14	75.0%	21.4%
Hinglish call lines	124	22	100%	0%
Near-domain genuine	0	30	–	10.0%
Fresh set[†]	10	10	10/10	2/10
Spot-check kit	28 of 30 items correct			

Share flagged (AMBER or RED). [†]Calls over a simulated phone line.

from training (75.8% clear, 54.6% subtle), which bounds what the model generalises.

Our sets and unseen inputs. Table III reports the combined system on our own held-out sets. The fresh set was written after the system was frozen and used for nothing else: six calls spoken by voices absent from all training data, each also passed through a simulated phone line, and four messages per class. All ten scams were flagged. The genuine items flagged were people discussing a scam and a bank fraud alert read by a synthetic voice.

Audio end to end. On 59 spoken calls (32 scams) passed through the recogniser, the conversation model alone flags 66% of scams on clean audio, 62% over G.711 and 50% after loudspeaker replay plus G.711, with 7–11% of genuine calls at AMBER and at most 4% at RED. Here the keyword path did better on scams (88% over G.711): our English scam scripts are more formal than ICFD's bank-call English. The combined system has not yet been measured on this audio set.

Voice. Table IV explains why the voice check is weak evidence: equal error rates of 23–35% on realistic channels are too high to justify an alarm on their own.

IX. LIMITATIONS AND FUTURE WORK

Measured weaknesses. (i) Scams read politely from a script in English can stay below AMBER. (ii) People talking *about* a scam can trigger AMBER, occasionally RED. (iii) A genuine call in a synthetic voice can reach AMBER; the card then names the voice as the reason. (iv) Short calls rarely reach RED, since it requires evidence over two segments. (v) Recognition errors over a loudspeaker lower detection (Section VIII).

TABLE IV
VOICE-CLONE CHECK (SHIPPED INT8 MODEL)

Held-out set	Channel	Equal error rate
In-the-Wild	clean / G.711	23.2 / 24.6%
ReplayDF (replayed by speaker)	G.711	35.2%
Stock TTS vs. real Hindi	clean / G.711	10.5 / 17.5%

Deployment constraints. On the earpiece the microphone mostly hears the user, so the call must be on speaker, and some handsets' echo cancellation removes the caller even then. ICFD-31k is synthetic; performance on real recorded calls needs a pilot with consented data.

Not yet built. A full-screen block over payment apps (designed; today the call card and home screen warn), a small on-device language model as a third judge (prompt and parser written and tested, no runtime yet), three specified phone signals (screen sharing, long unknown video calls, 1600-series numbers), and the combined system on the server.

X. CONCLUSION

Countersign shows that impersonation scams can be detected on the victim's phone, in Indian languages, with an explanation the victim can read, and that the decisive step, bringing in the family, can be automated without sending the conversation anywhere. Judging the whole conversation rather than its keywords cut false warnings on genuine bank calls from one in two to about two in a hundred.

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